

Number Theory Problem Solving

- Gaurish

<https://codeforces.com/contest/222/problem/C>

C. Reducing Fractions

time limit per test: 2 seconds

memory limit per test: 256 megabytes

To confuse the opponents, the Galactic Empire represents fractions in an unusual format. The fractions are represented as two sets of integers. The product of numbers from the first set gives the fraction numerator, the product of numbers from the second set gives the fraction denominator. However, it turned out that the programs that work with fractions in this representations aren't complete, they lack supporting the operation of reducing fractions. Implement this operation and the Empire won't forget you.

Input

The first input line contains two space-separated integers n, m ($1 < n, m < 10^5$) that show how many numbers the first set (the

n m

a_1 a_2 ... a_n

b_1 b_2 ... b_m

$$\underline{\text{num}} = \prod a_i$$

$$\underline{\text{den}} = \prod b_i$$

→ simplify

$\sum a_i$

$\prod a_i$

↑ product

$$\frac{5}{10} \rightarrow \frac{1}{2} \quad \text{coprime}$$

$$\frac{15}{20} \rightarrow \frac{3}{4} \quad \text{coprime}$$

$$\begin{array}{|c|c|} \hline 5 & 3 \\ \hline 5 & 4 \\ \hline \end{array}$$

Common part
GCD

$$\frac{n}{d} \rightarrow \frac{n / \gcd(n, d)}{d / \gcd(n, d)}$$

Clear??

$\nearrow \underline{a_i}, \underline{b_i} \leq 10^7$

$\prod a_i \leq (10^7)^n \approx (10^7)^{10^5}$

we can't store it

$\times$

$$\begin{aligned} \rightarrow a &= p_1^{k_1} p_2^{k_2} \\ \rightarrow b &= p_1^{k_3} p_2^{k_4} \end{aligned}$$

$$\text{GCD} = p_1^{\min(k_1, k_3)} p_2^{\min(k_2, k_4)}$$

$$\left[P : \min(f_{p_1}, f_{p_2}) \text{ GCD} \right]$$

$$\begin{array}{ccc}
 \frac{5}{2} & \frac{2}{2} & \frac{10}{5 \cdot 2} \\
 \downarrow & \downarrow & \downarrow \\
 5^1 & 2^1 & 5^1 \cdot 2^1
 \end{array}
 \qquad
 \begin{array}{c}
 5 \cdot 2 = 10 \\
 \downarrow \\
 5^2 \cdot 2^2
 \end{array}$$

mp

$$\left[\begin{array}{c} 5 : [2] \\ 2 : [2] \end{array} \right]$$

$\nearrow$ num
 $\searrow$ denom

$$a_i, b_i \leq 10^7$$

↳ spf $\Rightarrow$ smallest prime factor

$$\text{↳ } O(\log_2(a_i)).$$

$$\text{↳ } O(n \cdot \log_2(b^7))$$

↓

$$\begin{array}{r} 5^3 2^2 \\ 1000 \quad 5 \quad 2^1 \\ \hline 5^2 2 \times 5 2^1 \\ 50 \quad 10 \end{array} \leftarrow 5^3 2^2$$

$$\begin{array}{r} 5^3 2^2 \\ 5^3 2^2 \end{array}$$

$$\boxed{5^3 2^2}$$

$$\begin{array}{ccccccc} 1 & \times & 1 & \times & 2 & & \\ 1 & \times & 1 & & & & \\ & & & & & \rightarrow & \frac{2}{1} \end{array}$$

$q_i \rightarrow$ prime factorisation

$$q_i \Rightarrow \underbrace{p_1^{k_1}}_{\text{prime}}, \underbrace{p_2^{k_2}}_{\text{prime}}, \underbrace{p_3^{k_2}}_{\text{prime}}$$

$$p_1 \stackrel{\text{is}}{=} x \leftarrow \boxed{\min(k_1, x)} \\ p_2 = y$$

$$p_3 = z$$

<https://codeforces.com/contest/1992/problem/F>

F. Valuable Cards

time limit per test: 4 seconds

memory limit per test: 512 megabytes

In his favorite cafe Kmes once again wanted to try the herring under a fur coat. Previously, it would not have been difficult for him to do this, but the cafe recently introduced a new purchasing policy.

Now, in order to make a purchase, Kmes needs to solve the following problem: n cards with prices for different positions are laid out in front of him, on the i -th card there is an integer a_i , among these prices there is no whole positive integer x .

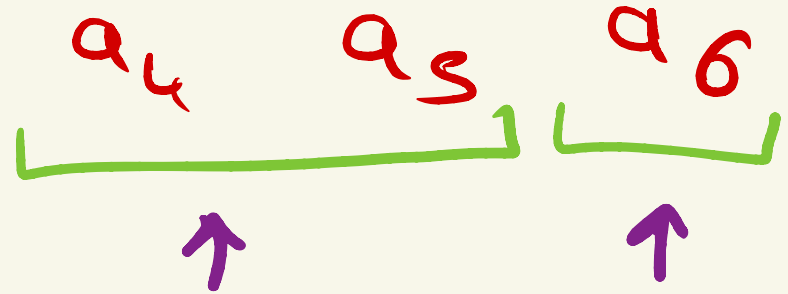
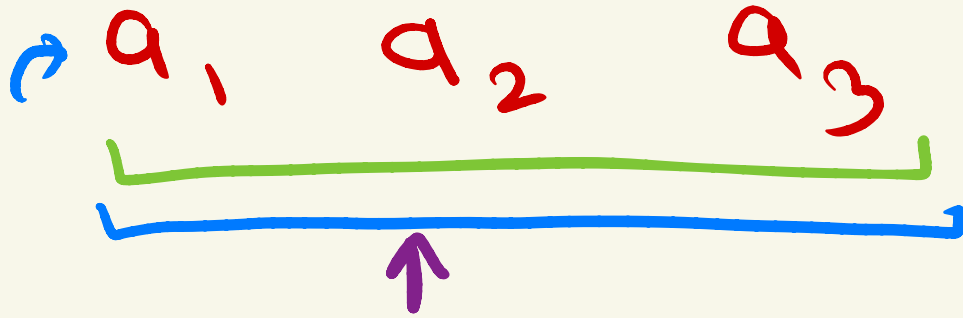
Kmes is asked to divide these cards into the minimum number of *bad* segments (so that each card belongs to exactly one segment). A segment is considered *bad* if it is impossible to select a subset of cards with a product equal to x . All segments, in which Kmes will divide the cards, must be *bad*.

Formally, the segment (l, r) is *bad* if there are no indices $i_1 < i_2 < \dots < i_k$ such that $l \leq i_1, i_k \leq r$, and $a_{i_1} \cdot a_{i_2} \dots \cdot a_{i_k} = x$.

Help Kmes determine the minimum number of *bad* segments in order to enjoy his favorite dish.

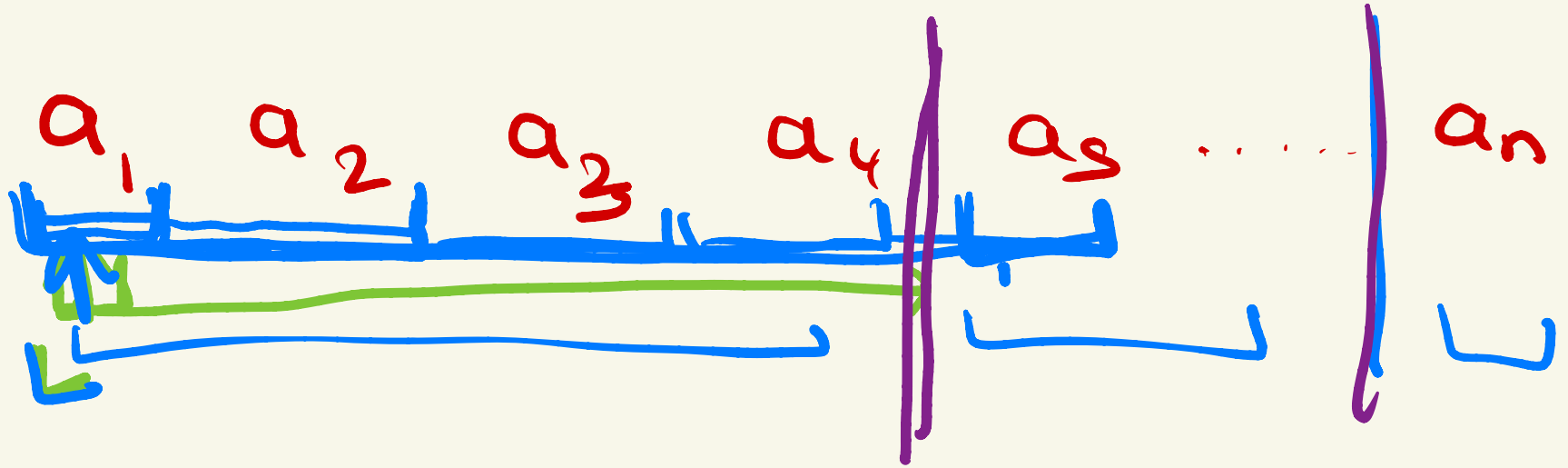
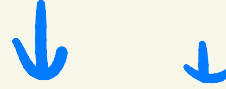
n x

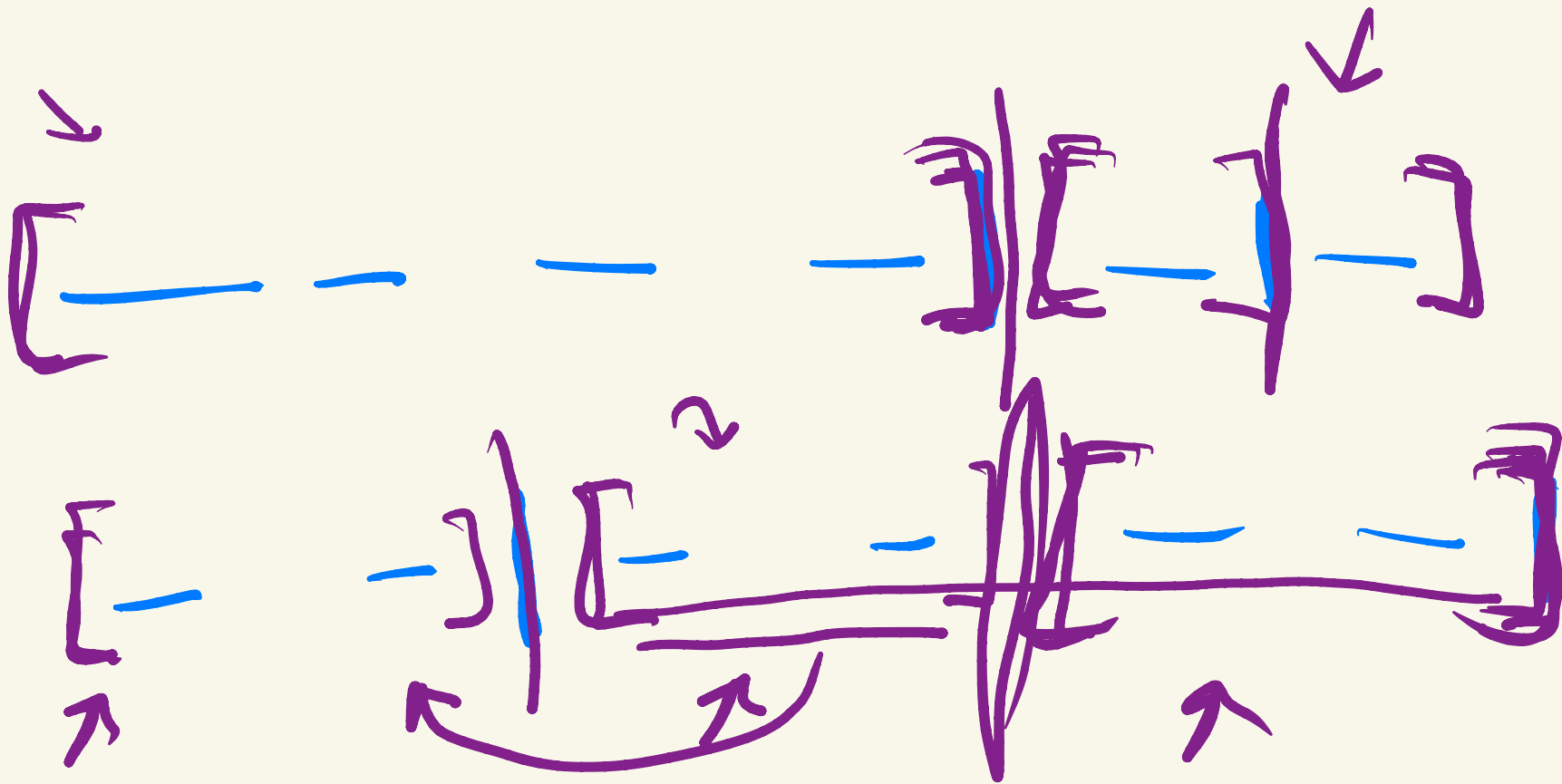
min number of
such segments

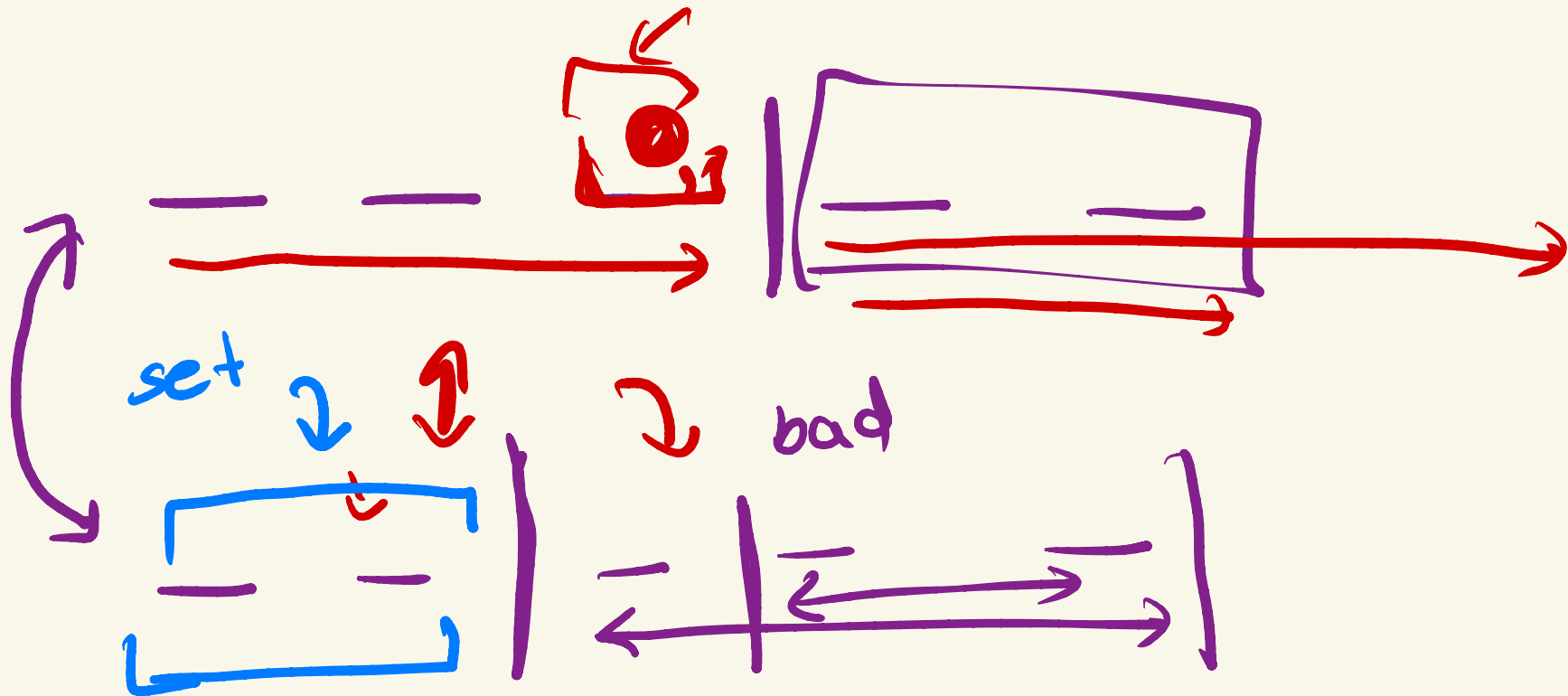


Its not possible to get
product of sub seq = x
in that subarray

$$\text{ans} = 1 + 1 + 1 = 3$$





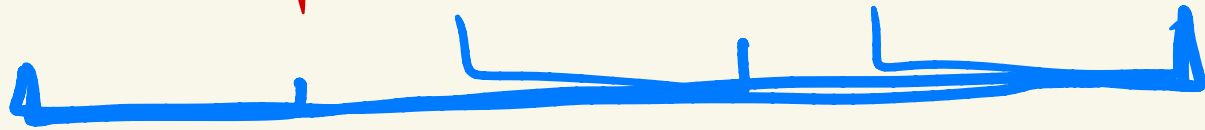


How to check? →

factors of x

↓ ↓ ↓

$$a_{i_1} \times a_{i_2} \times a_{i_3} = x$$



set <int> f ← $[F]$

↑ factors of a are possible

[]



Number factors

x

$\approx$

$\text{cbrt}(a)$

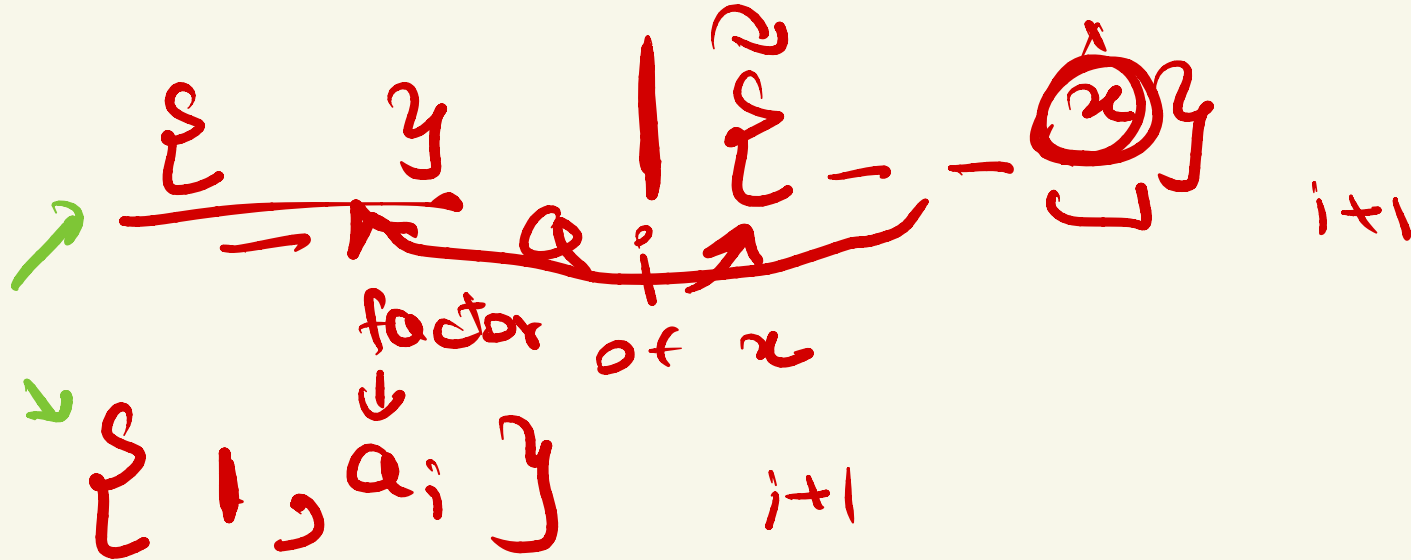
$\pm \epsilon$

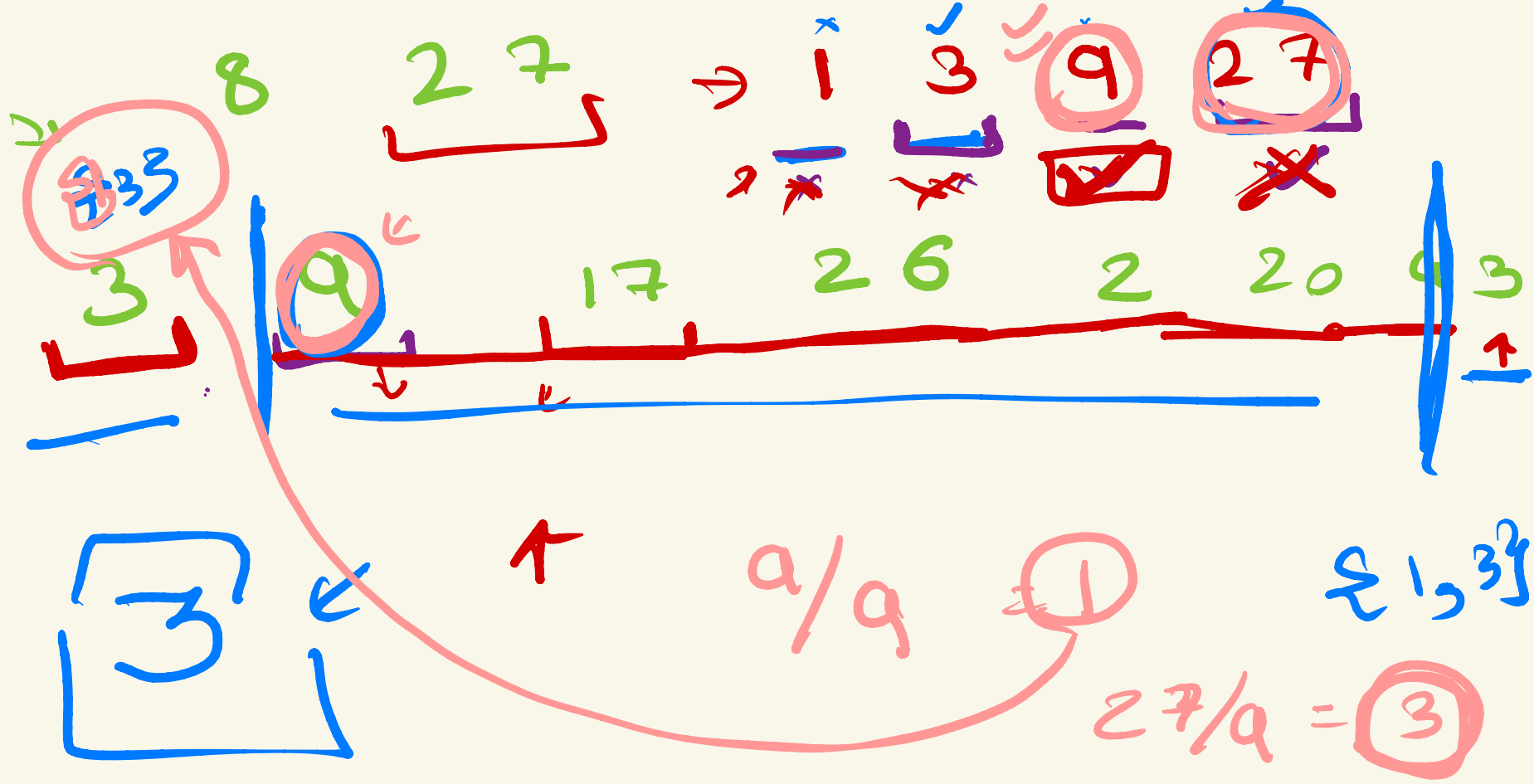
size of set $\leftarrow \chi_{cbrt}(n)$

$$\underbrace{10^8 \times 50}$$

$$\underbrace{\rightarrow \approx 47 \quad \boxed{50}}$$

we DONT want to make 2





$$n \times \text{cbrt}(n) \cdot \log(n)$$

↑

↑

30

17



~
~
~
 10^8

~
~
~
 $10^5 \times 1800$

~
~
~
 10^8

n^2

$$\rightarrow 10^8 \times 10^8 \times 12$$

$$= 10^{10} \times \underline{\underline{12}}$$

$$10^8 \approx 1 \text{ sec}$$

$$\frac{10^{10}}{10^8} = \boxed{100} \text{ sec}$$

factor = $\Sigma \textcircled{1}, 2^3$
 $\uparrow$ $\uparrow$

2

$\nearrow$

1

$\searrow$

2

$\searrow$

4

$2/2$

✓

$4/2$ $\rightarrow$

✓

<https://cses.fi/problemset/task/2182>

CSES Problem Set

Divisor Analysis

TASK

SUBMIT

RESULTS

STATISTICS

TESTS

QUEUE

Time limit: 1.00 s **Memory limit:** 512 MB

Given an integer, your task is to find the number, sum and product of its divisors. As an example, let us consider the number 12:

- ✓ the number of divisors is 6 (they are 1, 2, 3, 4, 6, 12)
- ✓ the sum of divisors is $1 + 2 + 3 + 4 + 6 + 12 = 28$
- the product of divisors is $1 \cdot 2 \cdot 3 \cdot 4 \cdot 6 \cdot 12 = 1728$



Since the input number may be large, it is given as a prime factorization.

$$n = p_1^{k_1} \cdot p_2^{k_2} \dots$$

$$\text{num Divisors} = \prod (k_i + 1)$$

$$\rightarrow (k_1 + 1)(k_2 + 1)$$

$$10 = 2^1 \cdot 5^1 \rightarrow (1+1)(1+1)$$

$$= 2 \times 2 = 4$$

$$1 + s + 2 + 10$$

$$(2^0 + 2^1)(s^0 + s^1)(7)$$

$$\rightarrow 2^0 s^0 + 2^0 s^1 + 2^1 s^0 + 2^1 s^1$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$1 \quad s \quad 2 \quad 10$$

$$n = \underline{p_1^{k_1} p_2^{k_2} p_3^{k_3}}$$

$$(\underline{p_1^0} + \underline{p_1^1} + \dots + \underline{p_1^{k_1}}) \times (p_2^0 + p_2^1 \dots p_2^{k_2})$$

$$\times (p_3^0 + p_3^1 \dots p_3^{k_3})$$

Sum of GP

previous
class

Product

$$10 \Rightarrow 1 \cdot 2 \cdot 5 \cdot 10$$

$$\hookrightarrow 2^0 5^0 \cdot 2^1 \cdot 5^0 \cdot 2^1 5^1 \cdot 2^1 5^1$$

$$\underbrace{2^2}_{1} s^1$$

$$\{1, s\} \rightarrow 2 \quad \frac{\pi(k;+1)}{I}$$

$$1.5$$

$$1.5 \cdot (1.2) \cdot (5.2) \cdot (1.2) \cdot (5.2) \cdot d^0$$

$$1.5 \cdot [1.5]^2 \cdot (2^1 \cdot 2^2)^2$$

$$\text{prev} =$$

$$\left[\begin{array}{l} \downarrow k_i + 1 \\ \frac{p}{p_{old}} \cdot \left(\text{prime}_i \right) \end{array} \right] \rightarrow k_i(k_i+1)/2$$

numDiv_{old}

$$\underline{2^2} \underline{5^1}$$

$$\text{num Div} = \underline{3}$$

$$p = 18$$

$$p_{\text{new}} = (8)^2 \cdot (\underline{5^1})^3$$

$$\underline{8^2} \times (\underline{5^1})^3 = \frac{64 \times 125}{\quad}$$

$$m \times 2 = \underline{6}$$

$$a^{(b^{\phi(\text{prime})})}$$

$$\% 10^9 + 6$$

$$\left[\begin{array}{l} (3 \times 5 \times 7) \% 6 = 6 \\ ((3 \times 5) \% 6 \times 7) \% 6 \end{array} \right]$$

$2^2 s^1$

$2^0 s^0$

$2^1 s^0$

$2^2 s^0$

$2^0 s^1$

$2^1 s^1$

$2^2 s^1$

]

6